

CSE-353: System Analysis and Design

Section-A — Complete Solutions (2019 – 2024)

Compiled for CUET Level-3 Students

This document contains detailed solutions to all Section-A questions from the CSE-353 examinations held between 2019 and 2024. Each solution is written concisely yet comprehensively to help students understand the underlying concepts of System Analysis and Design.

Q1(a) Why do we need a computer-based information system? Give an example of data and information obtained by processing data.

Need for a Computer-Based Information System (CBIS):

- **Speed & Accuracy:** Computers process millions of records in seconds with negligible error.
- **Storage:** Vast amounts of data can be stored and retrieved instantly.
- **Decision Support:** Managers get timely, relevant summaries for better decisions.
- **Automation:** Routine tasks (payroll, billing) are automated, freeing human effort.
- **Integration:** Different departments share a common database, eliminating redundancy.
- **Competitive Advantage:** Real-time data allows firms to respond quickly to market changes.

Example — Data vs. Information:

Data (raw facts): Student ID: 2001001, Marks: 78, 85, 60, 90, 55.

Processing: Sum = 368, Total marks = 500, Percentage = 73.6%, Grade = B+.

Information (processed, meaningful): "Student 2001001 has scored 73.6% and is awarded grade B+, qualifying for a merit scholarship."

Q1(b) Describe the qualities of information. What are the functional management areas in a large university?

Qualities of Information (TARCCA):

- **Timely:** Available when needed — stale data is worthless for decisions.
- **Accurate:** Free from errors; reflects reality correctly.
- **Relevant:** Pertains to the decision at hand; irrelevant data creates noise.
- **Complete:** Covers all necessary aspects; missing data leads to wrong decisions.
- **Concise:** Presented without unnecessary detail so the user can absorb it quickly.
- **Accessible:** Can be retrieved easily by authorised users.

Functional Management Areas in a Large University:

- Academic Affairs (Teaching & Learning)
- Research & Development
- Student Affairs & Admissions
- Finance & Accounts
- Human Resource Management (Faculty & Staff)
- Library & Information Services
- Examination & Results
- Estate & Physical Plant Management
- Alumni & External Relations

Q1(c) Strategic information for a garment buying house — example and justification.

A **buying house** acts as an intermediary between overseas buyers and local garment manufacturers.

Strategic Information Examples:

- **Market Trend Analysis:** Which product categories (e.g., knitwear vs. woven) are growing in the EU/US markets — helps decide which factories to empanel.
- **Competitive Pricing Intelligence:** Average FOB prices quoted by rival buying houses — informs pricing strategy.
- **Supplier Performance KPIs:** On-time delivery rate, defect rate, compliance audits — decides long-term supplier partnerships.
- **Trade Policy Changes:** GSP benefits, quota status — affects country-of-origin strategy.

Justification: Strategic information is long-range, summarised, and used by top management to set organisational goals. For example, knowing that knitwear demand from Germany will rise 20% next year allows the CEO to negotiate capacity with knit factories six months in advance — this cannot be derived from day-to-day operational data alone.

Q1(d) Relationship between analysis and design. Can a system be designed without analysis?

Relationship: Analysis *understands* the current system and the problem; design *creates* a solution. Analysis feeds requirements into design; design refines understanding back into analysis (iterative). They are complementary, not sequential in modern practice.

Can we design without analysis? — No (in practice):

- Without knowing user requirements, the designer has no target.
- Risk of building the wrong system — costly rework.
- Non-functional requirements (performance, security) are discovered only during analysis.

Theoretical exception: In very small, trivial problems a developer might skip formal analysis and code directly (cowboy coding), but this invariably leads to maintenance problems at scale. Even agile sprints include a mini-analysis phase (story elaboration) before each design/coding iteration.

Q2(a) Why organizations prefer a centralised database over independent files. Problems solved by DBMS.

Advantages of Centralised Database:

- **Data Consistency:** Single source of truth — updates reflect everywhere instantly.
- **Reduced Redundancy:** Data stored once instead of duplicated across files.
- **Data Integrity:** Constraints (PK, FK) enforced at the database level.
- **Concurrent Access:** Multiple users/applications access data simultaneously with locking.
- **Security:** Centralised access control — roles, privileges.
- **Backup & Recovery:** One location to back up; transaction logs allow point-in-time recovery.

Problems in Traditional File Systems Solved by DBMS:

File System Problem	DBMS Solution
Data redundancy & inconsistency	Normalisation + single storage
Difficult data access (custom programs)	SQL — ad-hoc queries
Data isolation (multiple formats)	Unified schema
Integrity problems	Declarative constraints
Concurrent-access anomalies	Transaction management (ACID)
Security problems	Role-based access control

Q2(b) Cost-Benefit Analysis — NPV for 5-year project.

Given: Initial Investment = Tk. 3,00,000; Benefits: 70,000 | 90,000 | 80,000 | 60,000 | 40,000; $r = 8\%$

Net Present Value (NPV) = $\sum B_t/(1+r)^t - C$

Year (t)	Benefit (Tk)	PV Factor ($1/1.08^t$)	PV of Benefit (Tk)
1	70,000	0.9259	64,813
2	90,000	0.8573	77,160
3	80,000	0.7938	63,504
4	60,000	0.7350	44,100
5	40,000	0.6806	27,224
Total PV			2,76,801

$NPV = 2,76,801 - 3,00,000 = -23,199 \text{ Tk.}$

Decision: Reject the project because $NPV < 0$, meaning the project does not cover its cost of capital at 8% and destroys value.

Q2(c) Features of effective output design and its impact on UX and decision-making.

Features of Effective Output Design:

- **Relevance:** Show only information the user needs — avoid information overload.
- **Timeliness:** Output must be available when decisions are being made.
- **Accuracy:** Data must be correct and up-to-date.
- **Clarity:** Use readable fonts, consistent layout, proper labels and units.
- **Appropriate Medium:** Screen, print, audio, or dashboard — match the use case.
- **Exception Reporting:** Highlight anomalies rather than listing all records.
- **Interactivity:** Allow filtering, drill-down for ad-hoc queries.
- **Aesthetics:** Logical grouping, white space, colour coding aid comprehension.

Impact: Well-designed output reduces cognitive load, so decision-makers understand the situation at a glance, spot trends faster, make fewer errors, and act more confidently. Poor output design wastes time, causes misinterpretation, and undermines trust in the system.

Q2(d) Three advantages of CASE Tools in software development.

- **1. Automated Documentation:** CASE tools generate diagrams (DFD, ER, UML), data dictionaries, and reports automatically, keeping documentation synchronised with the design and saving hours of manual work.
- **2. Consistency & Error Checking:** Tools enforce naming conventions, detect dangling data flows, unbalanced DFDs, and duplicate entity names — catching design errors before coding begins.
- **3. Productivity & Reusability:** Drag-and-drop interfaces, code-generation from models, and reusable component libraries speed up development significantly and reduce human typing errors.

Q3(a) Analyst preparation for an interview. Advantages of questionnaires.

Analyst Preparation Before an Interview:

- Study background documents, org charts, current system manuals, and existing reports.
- Identify the interviewee's role and what domain knowledge they hold.

- Prepare a structured list of open-ended and closed-ended questions.
- Arrange a suitable time and place, and inform the interviewee in advance.
- Bring recording tools (notepad/recorder) with permission.
- Plan for follow-up probing questions on expected sensitive areas.

Advantages of Questionnaires in Fact-Gathering:

- **Wide Reach:** Can collect data from hundreds of users simultaneously.
- **Anonymity:** Respondents answer more honestly about sensitive issues.
- **Consistency:** Every respondent answers the same questions — data is comparable.
- **Cost-effective:** No interviewer travel; digital forms reduce paper cost.
- **Time-flexible:** Respondents complete at their convenience.

Q3(b) How are goals determined? How is SRS developed?

How Goals are Determined:

- Interview senior management to understand organisational mission and strategy.
- Review strategic plans, budgets, and existing problem reports.
- Conduct JAD (Joint Application Design) sessions with stakeholders.
- Distinguish main goals (broad outcomes) from sub-goals (specific, measurable objectives).
- Prioritise goals using MoSCoW (Must/Should/Could/Won't) or weighted scoring.

How SRS is Developed:

- **Feasibility Study:** Confirm the project is viable.
- **Requirements Elicitation:** Interviews, questionnaires, observation, document analysis.
- **Requirements Analysis:** Model requirements using DFDs, ERDs, use-cases.
- **Requirements Specification:** Write formal descriptions — functional (what the system does) and non-functional (performance, security, reliability).
- **Review & Validation:** Stakeholders sign off; ambiguities resolved.
- **Baseline & Version Control:** SRS is baselined for change control.

Q3(c) Physical and Logical DFDs for the vendor delivery scenario.

Scenario: A company receives items from vendors with delivery notes; receiving office checks note against order; discrepancies reported to purchase office; items + receipt note sent to quality inspection.

Context Diagram (Level-0):

External entities: **Vendor** → [System] → **Purchase Office, Quality Inspection Office**

Inputs: Items + Delivery Note. Outputs: Discrepancy Report, Items Received Note.

Physical DFD (Level-1) — reflects actual people and places:

Vendor → (Items + Delivery Note) → *Receiving Clerk*

Receiving Clerk checks against *Purchase Order File*

If discrepancy → *Purchase Office*

Receiving Clerk → (Items + IRN) → *Quality Inspection Clerk*

Logical DFD (Level-1) — reflects what is done, not who does it:

Process 1.0: **Receive Delivery** — inputs: Delivery Note, Purchase Order. Output: Verified Receipt or Discrepancy Report.

Process 2.0: **Report Discrepancy** — inputs: Discrepancy details. Output: Discrepancy Report to Purchase Office.

Process 3.0: **Forward to Quality Inspection** — inputs: Items + IRN. Output: Items + IRN to Quality Inspection Office.

Data Store: D1 — Purchase Orders; D2 — Items Received Notes.

Q3(d) How does inflation affect payback period?

Inflation reduces the **purchasing power of money** over time.

- **In nominal terms:** If benefits are fixed in nominal Tk., their real value falls each year due to inflation, so the payback period effectively lengthens.
- **In real terms:** Analysts use the real interest rate (Fisher equation: $r_{\text{real}} \approx r_{\text{nominal}} - \text{inflation}$) when discounting. Higher inflation → higher nominal rate needed → lower PV of future benefits → longer payback.
- **Practical effect:** A project estimated to break even in 3 years at 0% inflation may take 4+ years when 8% annual inflation is factored in.
- **Conclusion:** Inflation always makes payback period longer and makes projects look less attractive, which is why analysts use discounted payback period with the real discount rate.

Q4(a) Decision Table for bank deposit interest policy.

Conditions: Amount $\geq$ 5000? Period $\geq$ 3 years?

Condition	Rule 1	Rule 2	Rule 3
Amount $\geq$ Tk. 5000?	Y	Y	N
Period $\geq$ 3 years?	Y	N	-
Interest Rate = 15%	X		
Interest Rate = 12%		X	
Interest Rate = 10%			X

'-' means 'Don't Care' (irrelevant condition). 'X' marks the action taken.

Q4(b) Data dictionary reasons. Payback period calculation.

Reasons for Compiling a Complete Data Dictionary:

- Provides a precise, shared definition of every data element — eliminates ambiguity.
- Enforces consistency in naming and formats across the entire system.
- Supports documentation and maintenance — new developers understand data quickly.
- Helps analysts verify completeness of DFDs (every flow defined).
- Facilitates database design (attributes map to table columns).

Payback Period Calculation:

Cost: Tk. 1,00,000 install + Tk. 10,000/month = Tk. 1,20,000/year recurring.

Net annual benefit = Tk. 1,50,000 – Tk. 1,20,000 = **Tk. 30,000/year**.

Discount rate = 12% → Discounted payback:

Year	Net Benefit (Tk)	PV Factor @12%	PV (Tk)	Cumulative PV (Tk)
0 (install)	-1,00,000	1.000	-1,00,000	-1,00,000

Year	Net Benefit (Tk)	PV Factor @12%	PV (Tk)	Cumulative PV (Tk)
1	30,000	0.8929	26,787	-73,213
2	30,000	0.7972	23,916	-49,297
3	30,000	0.7118	21,354	-27,943
4	30,000	0.6355	19,065	-8,878
5	30,000	0.5674	17,022	+8,144

Payback occurs between Year 4 and Year 5.

Exact payback $\approx 4 + (8,878/17,022) \approx 4.52$ years (≈ 4 years 6 months).

Q4(c) Must an operationally feasible solution also be technically feasible?

Not necessarily, but in practice the two must coexist for a project to succeed.

Operational Feasibility asks: Will users accept and use the system? Does it fit organisational workflow?

Technical Feasibility asks: Can it be built with available/acquirable technology?

Example 1 — Operationally feasible but NOT technically feasible:

A hospital wants real-time MRI image analysis by AI at the point of care. Staff are enthusiastic (operational ✓), but the required GPU compute is unavailable in a rural clinic and internet bandwidth is insufficient (technical ✗). The solution cannot be deployed as envisioned.

Example 2 — Technically feasible but NOT operationally feasible:

A sophisticated ERP system is technically buildable, but factory workers lack computer literacy and resist the change (operational ✗). The system would be abandoned.

Conclusion: Both feasibilities are necessary conditions. An organisation must confirm technical feasibility before committing design resources, and operational feasibility before expecting adoption.

Q4(d) Leveling rules of DFD.

DFD Leveling Rules:

- **Consistency (Balancing):** The inputs and outputs of a parent process must exactly match the inputs and outputs of its child DFD. No data flow can appear in the child if it did not exist in the parent.
- **Numbering:** Context diagram is Level 0. Level 1 processes are numbered 1, 2, 3...; their children 1.1, 1.2...; grandchildren 1.1.1, 1.1.2... and so on.
- **Decompose only when necessary:** A process should be decomposed further only if it still contains multiple logical sub-processes. A 'primitive' (leaf) process is a single, cohesive action.
- **Avoid mini-specs at non-primitive levels:** Process descriptions (mini-specs / structured English) are written only for primitive processes.
- **Data stores:** A data store referenced in a child DFD that was not shown in the parent is acceptable only if it is a local store used solely within that parent process.
- **External entities:** May be repeated across diagrams for clarity, but represent the same real-world entity.

Q1 Give an example of incomplete information. Why does an organisation need a CBIS?

Incomplete Information Example: A sales report showing total revenue as Tk. 50 lakh but not indicating which products, regions, or time periods generated it — the manager cannot decide where to invest or cut.

Why an Organisation Needs a CBIS:

- Handles huge volumes of transactions (orders, payroll, inventory) automatically.
- Provides accurate, timely summaries for management decision-making.
- Reduces human error and paper-based delays.
- Enables information sharing across departments through a central database.
- Supports strategic planning through trend analysis and forecasting.
- Complies with statutory/regulatory reporting requirements efficiently.

Q1 Statutory and strategic information for Chittagong Elevated Expressway. Example and justification.

Statutory Information (required by law):

- Environmental compliance reports submitted to the Department of Environment.
- Financial audit reports for the Ministry of Finance.
- Safety inspection certifications from BRTA/relevant authority.
- Land acquisition records with compensation proof.

Strategic Information (long-range, top management):

- **Traffic volume projections for 10–20 years** — determines toll pricing and lane capacity planning.
- **Cost-overflow analysis and revised completion timeline** — informs decisions about additional funding or phased opening.
- **Economic impact study** — justifies future extensions or complementary projects.

Justification: Strategic information is forward-looking and aggregated. Traffic projections, for instance, come from historical data combined with urban growth models — not from daily vehicle counts alone. Top management uses it to decide whether to expand to a 6-lane expressway or add entry/exit ramps, decisions that cannot wait for daily operational data.

Q1 Why are organisations divided into departments? Primary functions of Research Design and Development management in a university.

Why Organisations are Divided into Departments:

- **Specialisation:** Staff develop deep expertise in one function (finance, HR, etc.).
- **Accountability:** Clear responsibility and performance measurement.
- **Coordination:** Structured communication channels avoid chaos.
- **Efficient Resource Allocation:** Budgets assigned per department based on need.
- **Scalability:** New functions added as new departments without restructuring the whole org.

Primary Functions of Research, Design & Development Management in a University:

- Identifying research priorities aligned with national/global needs.
- Managing research grants and funding from government and industry.
- Facilitating faculty and student research projects, publications, and patents.

- Technology transfer — commercialising innovations through spinoffs or licensing.
- Maintaining research labs, equipment, and safety compliance.
- Collaborating with industry and international institutions for joint R&D.;

Q1 Qualities of Information.

(See 2024 Q1(b) answer above — qualities are identical: Timely, Accurate, Relevant, Complete, Concise, Accessible.)

Additional qualities often cited:

- **Verifiable:** Source traceable and cross-checkable.
 - **Economical:** Cost of producing the information should not exceed its value.
 - **Flexibility:** Useful for more than one purpose.
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Q2 Why can't managers design their own system?

Reasons Managers Cannot Design Systems:

- **Lack of Technical Skills:** System design requires knowledge of DBMS, networking, UI design, programming — skills outside a manager's domain.
- **Biased View:** Managers see their own department's needs; a system analyst takes a holistic, cross-departmental view.
- **Time Constraints:** Managers have operational responsibilities; system design is a full-time analytical task.
- **Methodology Gap:** Analysts use structured methodologies (SDLC, DFDs, ER modelling); managers are typically untrained in these.
- **Communication Barrier:** Analysts bridge between user language and technical language — managers often cannot translate their needs into system specifications.
- **Objectivity:** An external analyst can challenge existing practices objectively; a manager is invested in the status quo.

Q2 Stages of system life cycle. Possible sources of information for analysing CSE dept., CUET.

System Life Cycle Stages:

- 1. Preliminary Investigation (Feasibility Study)
- 2. Systems Analysis (Requirements Gathering)
- 3. Systems Design (Logical + Physical Design)
- 4. Systems Development (Programming & Testing)
- 5. Systems Implementation (Conversion & Training)
- 6. Systems Evaluation & Maintenance

Possible Sources of Information for Analysing CSE Dept., CUET:

- Interviews with Head of Department, faculty, students, and administrative staff.
- Examination of existing documents: course schedules, result sheets, admission forms.
- Observation of daily departmental workflows (lab usage, exam process).
- Questionnaires distributed to students and staff.
- Review of annual reports, accreditation documents, and existing software systems.
- Group discussions / JAD sessions with key stakeholders.

Q2 Distinguish between on-line and off-line data entry. Roles and tasks for a system analyst of CUET.

Aspect	On-Line Data Entry	Off-Line Data Entry
Definition	Data entered directly into live system via terminal.	Data collected first (paper/batch), then entered later
Timing	Real-time; immediate effect	Delayed; processed in batches
Validation	Immediate feedback on errors	Errors found later in batch processing
Examples	ATM transactions, web forms	Mark sheets collected daily, uploaded weekly
Cost	Higher (live connectivity needed)	Lower (can use simple forms)

Roles and Tasks for a System Analyst at CUET:

- Requirements gathering from faculty, students, and administration.
- Analysing existing manual or semi-automated systems (library, exam, payroll).
- Designing new system specifications and presenting proposals to management.
- Coordinating with programmers during development.
- Testing and validating the system before deployment.
- Training users and creating documentation.

Q3 Tools for a system analyst of a large superstore. Guidelines for a good interview.

Tools Used by a System Analyst:

- **Fact-Finding Tools:** Interviews, questionnaires, observation, document review.
- **Modelling Tools:** DFD (Data Flow Diagram), ERD (Entity-Relationship Diagram), UML diagrams.
- **Process Description Tools:** Structured English, Decision Tables, Decision Trees.
- **CASE Tools:** e.g., Visual Paradigm, Rational Rose, Enterprise Architect.
- **Project Management:** Gantt charts, PERT/CPM networks.
- **Prototyping Tools:** Mockup/wireframe tools (Figma, Balsamiq).

Guidelines for a Good Interview:

- Prepare questions in advance; share an agenda with the interviewee.
- Start with easy, rapport-building questions before sensitive ones.
- Use a mix of open-ended (why, how) and closed (yes/no) questions.
- Listen actively; take notes or record (with permission).
- Avoid leading questions that bias the answer.
- Clarify ambiguous answers immediately.
- Summarise and confirm key points at the end.
- Send a follow-up summary to the interviewee for verification.

Q3 How are goals determined? Tangible and intangible benefits examples.

How Goals are Determined:

- Identify the problem or opportunity triggering the project.
- Interview senior management to understand strategic direction.
- Analyse the gap between current system performance and desired performance.
- Break mission into main goals, then decompose into measurable sub-goals (SMART criteria).

Tangible Benefits (measurable in Tk.):

- Reduction in paper and printing costs by Tk. 50,000/year.
- Decrease in processing time from 3 days to 4 hours → labour savings Tk. 2 lakh/year.
- Reduction in inventory write-offs due to accurate stock management.

Intangible Benefits (real but hard to quantify):

- Improved customer satisfaction and loyalty.
- Better employee morale due to less repetitive manual work.
- Enhanced corporate image and competitiveness.
- Faster, more informed decision-making by management.

Q3 Distinguish among technical, operational, and economic feasibility.

Feasibility	Question Asked	Key Concerns	Example
Technical	Can it be built?	Technology availability, skills, infrastructure	Does CDE have servers to host the ERP?
Operational	Will it work in practice?	User acceptance, workflow fit, management support	Will clerks accept electronic leave forms?
Economic	Is it worth building?	Cost-benefit, NPV, ROI, payback period	Discounted savings of Tk.2L/yr justify Tk.5L investment?

Q4(a) Payback period — system costs Tk. 5 lakh + Tk. 4,000/month; benefit Tk. 1 lakh/year; rate 4%.

Annual recurring cost = $4,000 \times 12 = \text{Tk. } 48,000$.

Net annual benefit = $1,00,000 - 48,000 = \text{Tk. } 52,000$.

Undiscounted payback $\approx 5,00,000 / 52,000 \approx 9.6$ years.

Discounted payback at 4%:

Year	Net Benefit	PV Factor @4%	PV (Tk)	Cumulative PV
0	-5,00,000	1.000	-5,00,000	-5,00,000
1	52,000	0.9615	49,998	-4,50,002
2	52,000	0.9246	48,079	-4,01,923
3	52,000	0.8890	46,228	-3,55,695
4	52,000	0.8548	44,450	-3,11,245
5	52,000	0.8219	42,739	-2,68,506
6	52,000	0.7903	41,096	-2,27,410
7	52,000	0.7599	39,515	-1,87,895
8	52,000	0.7307	37,996	-1,49,899
9	52,000	0.7026	36,535	-1,13,364
10	52,000	0.6756	35,131	-78,233
11	52,000	0.6496	33,779	-44,454
12	52,000	0.6246	32,479	-11,975
13	52,000	0.6006	31,231	+19,256

Payback occurs between Year 12 and Year 13.

Exact: $12 + (11,975/31,231) \approx \text{12.38 years} (\approx \text{12 years 5 months})$.

Q4(b) Data dictionary — definition and reasons for compilation.

Data Dictionary: A centralised repository of definitions for all data elements in a system — names, descriptions, data types, lengths, valid values, sources, and relationships.

Reasons for Compiling a Complete Data Dictionary:

- Ensures all team members use consistent, unambiguous data definitions.
- Documents every data flow, data store, and process, making DFD validation possible.
- Facilitates database design — attributes become table columns.
- Aids maintenance — future developers understand data without asking the original team.
- Reduces data redundancy by identifying duplicate data elements.
- Supports system testing — test cases can be built from valid value ranges.

Q4(c) Decision Table and Tree for True Disk discount policy.

(i) Limited Entry Decision Table (LEDT):

Condition	R1	R2	R3	R4	R5	R6
Special order?	Y	N	N	N	N	N
Amount > Tk.10,000?	-	Y	Y	N	N	N
Amount Tk.5k–10k?	-	-	-	Y	Y	N
Order via web?	-	Y	N	Y	N	-
No discount	X					X
4% discount		X	X			
2% discount				X	X	
Extra 5% (web)		X		X		

(ii) Extended Entry Decision Table (EEDT):

Condition	Rule 1	Rule 2	Rule 3	Rule 4
Order type	Special	Normal	Normal	Normal
Amount (Tk)	Any	>10,000	5,000–10,000	<5,000
Web order?	Any	Yes/No	Yes/No	Yes/No
Base discount	None	4%	2%	0%
Web extra	None	+5% if web	+5% if web	N/A

(iii) Decision Tree (text representation):

Special Order? → YES → No discount.

Special Order? → NO →

Amount > Tk.10,000? → YES → Base 4% → Web? YES → Total 9% | NO → Total 4%.

Amount Tk.5k–10k? → YES → Base 2% → Web? YES → Total 7% | NO → Total 2%.

Amount < Tk.5,000? → YES → Base 0% → No extra discount.

Q4(d) Output method for Luke's scenario ('need it rarely, but very quickly when needed').

Luke's requirement: Infrequent access but instantaneous retrieval when needed.

Recommended Output Method: On-Demand Screen Display with Drill-Down (Online Enquiry System)

- **Why:** An online enquiry screen allows Luke to pull the data in seconds whenever needed, without waiting for a scheduled print run or report cycle.
- Data is always up-to-date (unlike pre-printed reports that go stale).
- Search/filter capability means Luke finds the specific record instantly.
- No storage problem — he doesn't need to file paper reports he rarely reads.
- If a hard copy is occasionally needed, a print-on-demand button handles that.

Alternative: A dashboard with KPI tiles and drill-down into details — Luke glances at the overview and clicks into specifics only when anomalies appear.

Q1(a) Non-arithmetic data processing example. Why CBIS?

Non-Arithmetic Data Processing: Sorting student names alphabetically, searching a customer record by name, merging two mailing lists, or generating a library catalogue — no mathematical operations involved.

(For CBIS reasons, see 2024 Q1(a) — same answer applies.)

Q1(b) How MIS differs from DSS. Qualities of information.

Aspect	MIS	DSS
Purpose	Routine reporting for operational/middle management	Supporting routine, semi-structured decisions
Users	Middle managers	Senior managers, analysts
Output	Pre-defined periodic reports	Ad-hoc analyses, what-if scenarios
Data	Internal operational data	Internal + external + historical
Flexibility	Low — fixed formats	High — interactive, model-driven
Example	Monthly sales summary	Market expansion scenario analysis

(Qualities of information: see 2024 Q1(b).)

Q1(c) Tactical and strategic information for CUET Health Center.

Tactical Information (medium-term, middle management):

- Monthly patient visit trends by department (OPD, pharmacy, dental).
- Medicine consumption rate vs. stock levels — order planning.
- Doctor and nurse utilisation rates — staffing adjustments.

Strategic Information (long-term, top management):

- 5-year disease prevalence trends — justifies new specialist clinics.
- Cost per patient vs. budget allocated — informs budget requests to CUET administration.
- Comparative health metrics with peer university health centres — benchmarking.

Justification: Strategic information is aggregated and future-oriented. Knowing that respiratory infections spike every November for 3 consecutive years allows the health centre director to proactively stock medicines and schedule additional doctors — a strategic decision that saves cost and improves service.

Q1(d) Relationship between analysis and design. Design without analysis?

(Same as 2024 Q1(d). Key point: Analysis → defines requirements; Design → creates solution satisfying requirements. Cannot design correctly without analysis.)

Q2(a) An excellent programmer is necessarily an excellent system analyst — justify.

Statement: FALSE

Reasons:

- **Different Skill Sets:** Programmers excel at coding logic and debugging; analysts excel at communication, problem-solving, and translating business needs to system specs.
- **Focus Difference:** Programmers think in terms of code; analysts think in terms of business processes and user requirements.

- **Communication:** Analysts must interview non-technical users and managers — a skill not required for programming.
- **Systems Thinking:** Analysts see the big picture across the organisation; programmers focus on a specific module.

However, a programmer with interpersonal and analytical skills *can* become a good analyst — programming experience helps understand technical feasibility.

Q2(b) How data will be processed for the functional requirements of a purchase department.

Purchase Department Functional Requirements and Data Processing:

- **Purchase Requisition:** Data input — item code, quantity, requesting department. Processing — validate against approved vendor list and budget. Output — approved/rejected requisition.
- **Purchase Order Creation:** Input — approved requisition. Processing — select vendor (lowest price, delivery time), generate PO number, update PO database. Output — PO sent to vendor.
- **Goods Receipt:** Input — delivery note, PO. Processing — match delivery note to PO, check quantities. Output — Goods Receipt Note (GRN) or discrepancy report.
- **Invoice Processing:** Input — vendor invoice, GRN. Processing — three-way match (PO + GRN + invoice). Output — payment authorisation or dispute notice.
- **Vendor Performance:** Input — delivery dates, quality reports. Processing — calculate KPIs. Output — vendor scorecard.

Q2(c) Stages of system life cycle. Sources of information for CSE dept. analysis.

(Same as 2023 Q2 answer on system life cycle stages and sources of information.)

Q2(d) Important points for designing a report format.

- Include a clear title, report date, and period covered.
- Identify the intended audience and tailor detail level accordingly.
- Organise data logically — most important information first (inverted pyramid).
- Use consistent column headings with units clearly labelled.
- Highlight exceptions or out-of-range values (bold, colour, asterisk).
- Include totals, subtotals, and summary statistics where applicable.
- Number pages and include a footer with data source.
- Avoid clutter — use white space, lines, and groupings for readability.

Q2(e) Advantages of questionnaires. Guidelines for good interviewing.

(See 2024 Q3(a) — same topics covered.)

Q3(a) Main goals vs. sub-goals. Payback period at 10% interest. Effect of inflation.

Main Goals vs. Sub-Goals:

Main goal: Broad, high-level objective aligned with organisational mission. E.g., 'Improve student academic performance at CUET.'

Sub-goals: Specific, measurable steps toward the main goal. E.g., 'Reduce failed courses by 20% in 2 years', 'Implement online tutoring by Dec 2025'.

Payback Period Calculation (10% interest):

Cost: Tk.1,00,000 + Tk.10,000×12 = Tk.1,20,000/yr recurring. Net benefit = Tk.1,50,000 – Tk.1,20,000 = Tk.30,000/yr.

Year	Net Benefit	PV@10%	PV (Tk)	Cumulative
0	–1,00,000	1.000	–1,00,000	–1,00,000
1	30,000	0.9091	27,273	–72,727
2	30,000	0.8264	24,793	–47,934
3	30,000	0.7513	22,539	–25,395
4	30,000	0.6830	20,490	–4,905
5	30,000	0.6209	18,627	+13,722

Payback $\approx 4 + (4,905/18,627) \approx 4.26$ years (≈ 4 years 3 months).

Effect of Inflation on Payback: Inflation erodes the real value of future cash flows. A higher inflation rate increases the nominal discount rate required, reducing the PV of future benefits and extending the payback period.

Q3(b) Operationally feasible vs. technically feasible. (Same as 2024 Q4(c).)

(See 2024 Q4(c) answer.)

Q4(a) Tools for system analyst of a large superstore. (Same as 2023 Q3.)

(See 2023 Q3 answer on tools.)

Q4(b) Context diagram. Leveling rules of DFD with example.

Context Diagram (Level-0 DFD): A single-process DFD showing the entire system as one process (circle), all external entities, and the data flows between them. It defines the system boundary.

Example: For a library system — External entities: Member, Librarian; Data flows in: Book request, Return request; Data flows out: Book, Receipt, Fine notice.

Leveling Rules (same as 2024 Q4(d)):

- Inputs/outputs of parent process must match child DFD (balancing rule).
- Sequential numbering: Level 0 → Level 1 (1,2,3...) → Level 2 (1.1,1.2...).
- Decompose only until primitive processes; write mini-specs only for primitives.
- Local data stores may appear in child DFDs without being in the parent.
- External entities may be repeated for diagram clarity.

Q4(c) Physical and Logical DFDs for vendor delivery scenario. (Same as 2024 Q3(c).)

(See 2024 Q3(c) answer.)

Q1(a) Why CBIS? Non-arithmetic data processing example.

(See 2024 Q1(a) and 2022 Q1(a).)

Q1(b) Why organisations divided into departments? Primary functions of Biman Bangladesh Airlines marketing dept. Qualities of information.

(Why departments: see 2023 Q1.)

Primary Functions of a Marketing Department — Biman Bangladesh Airlines:

- Market research — passenger demand analysis on domestic and international routes.
- Pricing strategy — competitive fare setting, seasonal promotions.
- Advertising and brand promotion — digital, TV, travel agency partnerships.
- Frequent flyer programme management — customer loyalty.
- Sales channel management — online booking, travel agents, corporate accounts.
- Customer relationship management — complaint handling, satisfaction surveys.

(Qualities of information: see 2024 Q1(b).)

Q1(c) Strategic, tactical, and operational information for Sheikh Kamal IT Business Incubator, CUET.

Operational Information (day-to-day):

- Daily visitor log and meeting room bookings.
- Startup team attendance and lab usage records.
- Equipment maintenance schedule.

Tactical Information (monthly/quarterly, middle management):

- Number of startups in each development phase.
- Monthly burn rate vs. funding received per startup.
- Mentor session frequency and outcomes.

Strategic Information (annual, top management):

- Number of startups that received external investment or revenue in past 3 years.
- Job creation impact — graduates employed by incubator companies.
- National ranking and comparison with peer incubators.
- Policy recommendations for expanding the incubator's capacity.

Q2(a) Personal qualities helpful to a system analyst. Why feasibility analysis is necessary.

Personal Qualities of a System Analyst:

- **Communication Skills:** Explain technical concepts to non-technical users.
- **Analytical Thinking:** Break complex problems into manageable parts.
- **Curiosity:** Ask probing questions to uncover hidden requirements.
- **Objectivity:** Evaluate options without bias.
- **Flexibility:** Adapt to changing requirements.
- **Interpersonal Skills:** Build rapport with users, managers, and developers.

- **Problem-Solving:** Creative, logical approach to finding solutions.

Why Feasibility Analysis is Necessary:

- Prevents wasting time and money on projects that are technically impossible or unaffordable.
- Identifies risks early when changes are cheap.
- Provides a rational basis for management to approve or reject the project.
- Ensures the proposed solution will actually be adopted by users (operational feasibility).

Q2(b) System Development Life Cycle — definition and stages.

SDLC: A structured process for planning, creating, testing, and deploying an information system. It ensures projects are completed on time, within budget, and meet user requirements.

Stages:

1. Preliminary Investigation — identify problem, determine feasibility.
2. Systems Analysis — detailed fact-finding, requirements documentation.
3. Systems Design — logical and physical design of the new system.
4. Systems Development — programming, unit testing, integration testing.
5. Systems Implementation — conversion (parallel/cutover), user training.
6. Systems Evaluation & Maintenance — post-implementation review, ongoing fixes.

Q2(c) Key roles and tasks performed by a system analyst.

- Investigate current system problems and user needs.
- Conduct feasibility studies.
- Gather and document system requirements (SRS).
- Create system models: DFDs, ERDs, use-case diagrams.
- Design input/output forms, reports, and database schemas.
- Liaise between users and programmers throughout development.
- Develop test plans and coordinate user acceptance testing.
- Write system documentation and train users.
- Evaluate and recommend improvements post-implementation.

Q2(d) How different departments interact. Example.

- Through shared databases — e.g., Sales and Inventory both access the product stock database.
- Through document flows — Purchase sends PO to vendor; Receiving sends GRN to Accounts.
- Through reports — Production sends daily output report to Planning for scheduling.
- Through meetings and approvals — HR approves recruitment requested by any department.

Example: In a garment factory, the Sales department receives an order → passes it to Planning → Planning checks Inventory for fabric → raises Purchase Requisition → Purchase Department orders fabric → Accounts processes payment. Each department's output is the next department's input.

Q3(a) Available tools for system design. Advantages.

Tools Currently Available:

- **DFD (Data Flow Diagrams):** Model data flows and processes.
- **ERD (Entity-Relationship Diagrams):** Model data structures.

- **UML Diagrams:** Use-case, sequence, class, activity diagrams for OO design.
- **Decision Tables & Trees:** Model complex logic/rules.
- **Structured English / Pseudocode:** Describe process logic precisely.
- **Flowcharts:** Visual program logic.
- **CASE Tools:** Enterprise Architect, Visual Paradigm, draw.io.
- **Prototyping Tools:** Figma, Balsamiq for UI mockups.

Advantages of Using These Tools:

- Visual representation improves communication with non-technical stakeholders.
- Errors in logic detected early before expensive coding starts.
- Documentation is auto-generated, staying in sync with design.
- Reusable components speed up design of similar systems.
- Standardised notation (UML, DFD symbols) understood globally by analysts.

Q3(b) Sources of information for CSE dept. analysis. Group discussions utility.

(Sources: see 2023 Q2.)

Are Group Discussions Useful in Information Gathering?

Yes, because:

- Multiple perspectives surface simultaneously — one person's comment triggers another's insight.
- Consensus on requirements is reached faster than individual interviews.
- Discrepancies between user views are identified and resolved in real time.
- Saves analyst time vs. conducting dozens of one-on-one interviews.

Caution: Dominant personalities may suppress minority views; the analyst must ensure all voices are heard.

Q3(c) How SRS is developed. (See 2024 Q3(b).)

(See 2024 Q3(b) answer.)

Q3(d) How are goals determined? Main goals vs. sub-goals with example.

(How goals are determined: see 2023 Q3.)

Main Goal vs. Sub-Goal — Example:

Main Goal: 'Improve academic performance of students at CUET.'

- Sub-goal 1: 'Reduce the failure rate in CSE-353 from 15% to 5% by 2025.'
- Sub-goal 2: 'Provide online mock exams for all Level-3 courses by Dec 2024.'
- Sub-goal 3: 'Assign an academic advisor to every student by Sept 2024.'

Main goals state the outcome; sub-goals define the specific, time-bound actions to achieve it.

Q4(a) Advantages of organising data using database approach.

- **Reduced Data Redundancy:** Data stored once; shared by many applications.
- **Data Consistency:** Update in one place reflects everywhere instantly.
- **Data Integrity:** Constraints enforced at DB level (PK, FK, CHECK).
- **Data Security:** Granular access control — users see only what they are permitted to.

- **Data Independence:** Application programs not affected by changes in data structure.
- **Concurrent Access:** Multiple users work simultaneously with transaction control (ACID).
- **Backup & Recovery:** Centralised backup; transaction logs for point-in-time recovery.
- **Ad-hoc Querying:** SQL allows flexible, non-programmed data retrieval.

Q4(b) NPV / Cost-benefit for Tk. 2 lakh project over 6 years at 10%.

Benefits: Yr1=50,000 | Yr2=80,000 | Yr3=90,000 | Yr4=70,000 | Yr5=50,000 | Yr6=30,000

Year	Benefit (Tk)	PV Factor @10%	PV (Tk)
1	50,000	0.9091	45,455
2	80,000	0.8264	66,115
3	90,000	0.7513	67,619
4	70,000	0.6830	47,811
5	50,000	0.6209	31,046
6	30,000	0.5645	16,934
Total PV			2,74,980

NPV = 2,74,980 – 2,00,000 = **+Tk. 74,980.**

Decision: Accept the project. NPV > 0 means the project generates value above the 10% required return.

Q4(c) Lou Cite's proposal — why not include all alternatives. What should appear.

i) Why not include all alternatives considered:

- Management has limited time — presenting every rejected option wastes their attention.
- Analysing all options in detail inflates the proposal with irrelevant material.
- It may confuse decision-makers and dilute focus on the recommended solution.
- Some alternatives may be confidential or based on incomplete analysis — presenting them could mislead.

ii) What should appear in the final proposal:

- **Problem statement** — concise description of the current system's deficiencies.
- **Objectives** — what the new system must achieve.
- **2–3 viable alternatives** that genuinely meet the objectives, with pros, cons, and cost estimates.
- **Recommended alternative** with full justification (cost-benefit, feasibility).
- **Implementation plan** — timeline, resources, milestones.
- **Anticipated benefits** — tangible and intangible.

Q4(d) Why system testing is required.

- **Detect Errors:** Find bugs in integrated modules that unit testing missed.
- **Validate Requirements:** Confirm the system does what the SRS specifies.
- **Performance Verification:** Check response times, throughput under expected load.
- **Security Testing:** Identify vulnerabilities before malicious actors do.
- **User Acceptance:** Ensure real users can operate the system comfortably (UAT).
- **Data Integrity:** Verify that data is not corrupted during processing.

- **Cost Reduction:** Fixing bugs post-deployment is 10–100× more expensive than fixing them in testing.

2020 — Level-3, Term-II

Q1(a) Why CBIS? Incomplete information example.

(CBIS reasons: see 2024 Q1(a).)

Incomplete Information Example: A payroll system reports total salary paid as Tk. 50 lakh but omits overtime breakdown, department-wise split, and tax deductions — Finance cannot reconcile accounts or file correct tax returns.

Q1(b) Strategic vs. tactical information. Examples for Biman Bangladesh Airlines.

Aspect	Strategic Information	Tactical Information
Time horizon	Long-term (3–5+ years)	Short-term (months to 1 year)
Users	Top management / Board	Middle management
Detail level	Highly summarised	Moderately detailed
Source	Mostly external + internal summaries	Primarily internal operational data
Purpose	Set direction, major policy	Monitor and control operations

Biman Bangladesh Airlines — Examples:

- **Strategic:** Which new international routes to open in the next 5 years based on demand forecasts.
- **Strategic:** Fleet replacement decision — Boeing 787 vs. Airbus A350.
- **Tactical:** Monthly load factor (% seats filled) by route — adjust pricing/scheduling.
- **Tactical:** Quarterly fuel consumption vs. budget — decide fuel hedging strategy.

Q1(c) Analysis vs. design relationship. Design without analysis?

(See 2024 Q1(d).)

Q1(d) Functional management areas in a large university.

(See 2024 Q1(b).)

Q2(a) Qualities of information. How departments interact.

(Qualities: see 2024 Q1(b). Departmental interaction: see 2021 Q2(d).)

Q2(b) Possible roles and tasks for a system analyst of CUET.

(See 2023 Q2 and 2021 Q2(c).)

Q2(c) Data processing for purchase department functional requirements.

(See 2022 Q2(b).)

Q2(d) Possible sources of information for analysing CSE dept., CUET.

(See 2023 Q2.)

Q3(a) Monthly statement with discount — Structured English and Decision Tree.

Assumed Discount Policy (typical scenario): Orders > Tk.10,000 get 10% discount; Tk.5,000–10,000 get 5%; < Tk.5,000 no discount. Payments within 10 days get additional 2%.

Structured English:

```
BEGIN GenerateMonthlyStatement
FOR EACH customer:
  Compute total_amount = SUM(all orders this month)
  IF total_amount > 10,000 THEN
    base_discount = total_amount × 10%
  ELSE IF total_amount >= 5,000 THEN
    base_discount = total_amount × 5%
  ELSE
    base_discount = 0
  END IF
  IF payment_within_10_days THEN
    early_discount = total_amount × 2%
  ELSE
    early_discount = 0
  END IF
  net_payable = total_amount – base_discount – early_discount
  Print statement with net_payable
END FOR
END
```

Decision Tree (text representation):

Total amount > Tk.10,000? → YES → 10% base → Paid in 10 days? YES → 12% total | NO → 10% total.
Total amount Tk.5k–10k? → YES → 5% base → Paid in 10 days? YES → 7% total | NO → 5% total.
Total amount < Tk.5,000? → YES → 0% base → Paid in 10 days? YES → 2% total | NO → 0% total.

Q3(b) Guidelines for good interviewing.

(See 2023 Q3.)

Q3(c) Context flow, physical and logical DFDs for CD/tape distribution company.

Assumed Scenario: Company receives orders from customers, processes them, picks items from warehouse, ships to customer, and sends invoice to accounts.

Context Diagram:

External entities: Customer, Supplier/Warehouse. System: Order Processing System.

Inputs: Customer Order. Outputs: Shipped Items, Invoice, Order Confirmation.

Physical DFD Level-1:

1.0 Order Clerk receives order → validates → enters into Order System.

2.0 Warehouse Picker retrieves CDs/tapes → packs → dispatches.

3.0 Accounts Clerk generates invoice → mails to customer.

Logical DFD Level-1:

1.0 Validate & Record Order → D1 Orders.

2.0 Pick & Ship Items (reads D1 Orders, D2 Inventory) → D3 Shipments.

3.0 Generate Invoice (reads D1, D3) → Customer.

Q3(d) Are excellent programmers necessarily excellent system analysts? (See 2022 Q2(a).)

(See 2022 Q2(a).)

Q4(a) Company X discount rules — LEDT, MEDT, and logical correctness check.

Assumed rules (typical Company X scenario):

Wholesale + Amount > Tk.1,000 → 20% discount.

Wholesale + Amount ≤ Tk.1,000 → 10% discount.

Retail + Amount > Tk.500 → 5% discount.

Retail + Amount ≤ Tk.500 → No discount.

(i) LEDT:

Condition	R1	R2	R3	R4
Wholesale?	Y	Y	N	N
Amount > Tk.1,000?	Y	N	-	-
Retail amount > Tk.500?	-	-	Y	N
20% discount	X			
10% discount		X		
5% discount			X	
No discount				X

(ii) MEDT:

Condition	Rule 1	Rule 2	Rule 3	Rule 4
Customer type	Wholesale	Wholesale	Retail	Retail
Order amount	>Tk.1,000	≤Tk.1,000	>Tk.500	≤Tk.500
Discount applied	20%	10%	5%	0%

(iii) Logical Correctness Check:

- Number of conditions = 2 (Wholesale? Y/N; Amount threshold).
- Maximum rules = $2 \times 2 = 4$. All 4 rules present → **No missing rules**.
- No two rules have identical condition entries → **No redundant rules**.
- No contradictions — each rule leads to a unique action.

Conclusion: Decision table is logically correct and complete.

Q4(b) Payback period — Tk.4 lakh install, Tk.4,000/month, Tk.1 lakh/yr benefit, 5% interest.

Net annual benefit = $1,00,000 - (4,000 \times 12) = 1,00,000 - 48,000 = \text{Tk.52,000}$.

Year	Net Benefit	PV@5%	PV (Tk)	Cumulative
0	-4,00,000	1.000	-4,00,000	-4,00,000
1	52,000	0.9524	49,524	-3,50,476
2	52,000	0.9070	47,164	-3,03,312
3	52,000	0.8638	44,918	-2,58,394
4	52,000	0.8227	42,780	-2,15,614
5	52,000	0.7835	40,742	-1,74,872
6	52,000	0.7462	38,802	-1,36,070
7	52,000	0.7107	36,956	-99,114
8	52,000	0.6768	35,194	-63,920
9	52,000	0.6446	33,519	-30,401
10	52,000	0.6139	31,923	+1,522

Payback $\approx 9 + (30,401/31,923) \approx 9.95$ years (≈ 10 years).

Q4(c) Cost-benefit analysis. Technical, operational, economic feasibility.

Cost-Benefit Analysis: A systematic comparison of the total costs of a proposed system against the total benefits it will generate, to determine whether the investment is justified. It includes NPV, payback period, ROI, and break-even analysis.

(Feasibility types: see 2023 Q3.)

Q4(d) Is it essential to have tangible benefits to justify an information system?

No, it is not essential.

Justification: Many information systems produce primarily intangible benefits that are difficult to quantify but are strategically critical.

Example: A CUET student grievance portal may cost Tk.5 lakh to build. Its direct financial savings are minimal. However, intangible benefits include: improved student morale, faster resolution of issues, reduced dropout rates, enhanced institutional reputation, and compliance with UGC governance requirements. These benefits justify the investment even though they cannot be expressed in Taka.

Management must exercise judgement to weigh intangible benefits against costs using qualitative scoring methods.

Q1(a) Distinguish between data and information. Non-arithmetic data processing example.

Aspect	Data	Information
Definition	Raw, unprocessed facts/figures	Processed, meaningful, contextual output
Form	Numbers, text, symbols	Reports, summaries, graphs
Usefulness	Not useful alone	Useful for decision-making
Example	Student marks: 78, 85, 60	Average = 74.3%, Grade = B

Non-Arithmetic Processing Example: Sorting a list of employee names alphabetically, searching a library catalogue by author, classifying products by category.

Q1(b) Qualities of information. How departments interact.

(See 2024 Q1(b) and 2021 Q2(d).)

Q1(c) Why CBIS? Strategic, tactical, operational information for an Electric Power Generation Corporation.

(CBIS reasons: see 2024 Q1(a).)

Electric Power Generation Corporation — Information Needs:**Operational:**

- Real-time power generation output (MW) per unit.
- Daily fuel consumption (coal/gas/oil) per plant.
- Equipment fault and maintenance alerts.

Tactical:

- Monthly generation capacity vs. demand — load-shedding schedule planning.
- Monthly fuel cost and variance from budget.
- Outage statistics — causes and duration by plant.

Strategic:

- 10-year demand forecast — where to build new plants.
- Renewable energy transition plan — solar/wind investment analysis.
- National grid integration strategy and cross-border power trading agreements.

Q1(d) Roles and tasks for a system analyst of CUET.

(See 2023 Q2 and 2021 Q2(c).)

Q1(e) Data mining. Differences between MIS and DSS.

Data Mining: The process of discovering patterns, correlations, anomalies, and useful insights from large datasets using statistical, machine learning, and database techniques. Examples: customer segmentation, fraud detection, demand forecasting.

(MIS vs. DSS: see 2022 Q1(b).)

Q2(a) System life cycle definition, stages, evaluation, and modification.

(SDLC definition and stages: see 2021 Q2(b).)

Evaluating and Modifying a System:

- **Post-Implementation Review:** Compare actual vs. planned performance metrics (speed, cost savings, user satisfaction).
- **User Feedback:** Surveys and interviews reveal usability issues.
- **Audit:** Check data accuracy, security compliance, and output quality.
- **Modification Process:** Log change requests → analyse impact → design change → test → deploy → document. Follow change control procedure to avoid unintended consequences.

Q2(b) Tools for system design and their advantages.

(See 2021 Q3(a).)

Q3(a) Cost-benefit analysis definition. Payback period — Tk.1 lakh, Tk.9,000/month, Tk.1.5 lakh/yr, 5%.

(Cost-benefit definition: see 2020 Q4(c).)

Net annual benefit = 1,50,000 – (9,000×12) = 1,50,000 – 1,08,000 = **Tk.42,000**.

Year	Net Benefit	PV@5%	PV (Tk)	Cumulative
0	–1,00,000	1.000	–1,00,000	–1,00,000
1	42,000	0.9524	40,000	–60,000
2	42,000	0.9070	38,094	–21,906
3	42,000	0.8638	36,280	+14,374

Payback $\approx 2 + (21,906/36,280) \approx 2.6$ years (≈ 2 years 7 months).

Q3(b) Data dictionary. Basic differences — Structured English, Decision Tables, Decision Trees.

(Data dictionary: see 2023 Q4(b).)

Aspect	Structured English	Decision Table	Decision Tree
Form	Pseudocode / prose	Matrix (rows/columns)	Tree / flowchart
Best for	Sequential, procedural logic	Complex conditions with many combinations	Hierarchical branching logic
Readability	High for technical users	High for complex rules at a glance	High for non-technical users
Completeness check	Manual review	Easy — count rules	Requires tracing all branches

Can we do the same task with all three? Yes. A bank's loan eligibility rule can be expressed as structured English IF-THEN-ELSE, or as a decision table with condition rows, or as a decision tree — all describe the same logic. Choice depends on complexity and audience.

Q3(c) University degree qualification — LEDT, MEDT, correctness check. Tangible vs. intangible benefits.

Assumed Rules: Student qualifies for degree if: passes all courses AND (Physics ≥ 50 OR Math ≥ 50) AND attendance $\geq 75\%$.

(a) LEDT:

Condition	R1	R2	R3	R4	R5
Passes all courses?	Y	Y	Y	Y	N
Physics ≥ 50 ?	Y	Y	N	N	-
Math ≥ 50 ?	Y	N	Y	N	-
Attendance $\geq 75\%$?	Y	Y	Y	-	-
Qualifies for degree	X	X	X		
Does NOT qualify				X	X

(b) MEDT:

Condition	Rule A	Rule B
Course result	All passed	Any failed
Science score	Physics ≥ 50 OR Math ≥ 50	-
Attendance	$\geq 75\%$	-
Outcome	Award degree	Deny degree

(c) Logical Correctness:

- With 4 binary conditions, max rules = 16. LEDT shows 5 representative rules covering all combinations via don't-care entries.
- No two rules have identical condition signatures $\rightarrow$ no redundancy.
- All outcomes covered $\rightarrow$ complete.

Tangible vs. Intangible Benefits: (see 2023 Q3.)

Q4(a) When to use SDLC, Agile, and Object-Oriented approaches.

SDLC (Waterfall):

- Use when requirements are well-understood, stable, and fully documented upfront.
- Suitable for large government projects, safety-critical systems (banking, payroll).
- Example: Designing a national ID card database for the government.

Agile Approach:

- Use when requirements are uncertain, likely to change, and user involvement is continuous.
- Suitable for web/mobile applications, startups, and customer-facing software.
- Example: Building an e-commerce app for a startup with evolving features.

Object-Oriented (OO) Approach:

- Use when the system involves complex real-world entities with behaviours and relationships.
- Suitable for systems with reusable components, GUI-heavy applications.
- Example: University ERP — Student, Course, Faculty objects with inheritance and polymorphism.

Q4(b) DFD symbols. Physical vs. logical DFD.

DFD Symbols:

- **Process (Circle / Rounded Rectangle):** A transformation that changes or moves data.
- **External Entity (Rectangle / Square):** Source or destination of data outside the system.
- **Data Store (Open Rectangle / Parallel lines):** A repository of data at rest.
- **Data Flow (Arrow):** Movement of data between entities, processes, or stores.

Aspect	Physical DFD	Logical DFD
Focus	How the system operates (who, where, what medium)	What the system does (functions, data)
Contains	Names of people, departments, documents, files	Abstract processes and data flows
Purpose	Model current (as-is) system	Model required (to-be) system
Example Process	"Sales Clerk fills order form"	"Record Customer Order"

Q4(c) Logical DFD for magazine subscription scenario.

Assumed Scenario: A magazine is published monthly. Customers subscribe, receive magazines, and renew/cancel subscriptions.

Context Diagram:

External entities: Subscriber, Post Office/Courier. System: Subscription Management System.

Inputs: New Subscription, Renewal, Cancellation. Outputs: Magazine, Invoice, Renewal Reminder.

Logical DFD Level-1:

1.0 **Process New Subscription** — Input: Subscriber details + payment. Output: Subscriber record (D1), Confirmation.

2.0 **Generate Monthly Dispatch List** — Input: D1 Active Subscribers. Output: Dispatch list.

3.0 **Send Renewal Reminders** — Input: D1 (subscriptions expiring this month). Output: Reminder notices.

4.0 **Process Renewal/Cancellation** — Input: Subscriber response. Output: Updated D1.

5.0 **Generate Invoice** — Input: D1. Output: Invoice to subscriber.

Q4(d) Physical and logical DFDs for library book issue, return, and train ticket reservation.

(i) Issuing a Book from Library:

Physical DFD: Member → (Book request) → Librarian Desk → checks Member Register + Book Catalogue → issues book → stamps Date Card.

Logical DFD: Process 1.0 Validate Member → Process 2.0 Check Book Availability (D1 Books, D2 Members) → Process 3.0 Issue Book → D3 Issued Books.

(ii) Returning a Book to Library:

Physical DFD: Member → returns book → Librarian checks date card → calculates fine (if any) → updates register.

Logical DFD: Process 1.0 Receive Return → Process 2.0 Calculate Fine (D3 Issued Books) → Process 3.0 Update Book Status (D1 Books, D3).

(iii) Train Ticket Reservation:

Physical DFD: Passenger → reservation form → Booking Clerk → checks Train Schedule file → allocates seat → prints ticket.

Logical DFD: Process 1.0 Check Availability (D1 Train Schedule, D2 Reservations) → Process 2.0 Reserve Seat → D2 Reservations → Process 3.0 Print Ticket → Passenger.

— End of Solutions —

This document was compiled for educational use by CUET CSE-353 students. All numerical calculations are shown step-by-step. Always cross-check with official course materials and textbooks.